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To produce 2 moles of HCl, 1 mole of chlorine gas is required

To produce 44.8 litres of HCl, 22.4 litres of chlorine gas are required.

∴ To produce 11.2 litres of HCl,

$$\frac{22.4 \text{ L } Cl_2}{44.8 \text{ L of HCl}} \times 11.2 \text{ L of HCl}$$

$$= \frac{1}{4}$$

= 5.6 litres of chlorine are required.

5. Calculate the percentage composition of the elements present in magnesium carbonate. How many kilogram of CO_2 can be obtained by heating 1 kg of 90 % pure magnesium carbonate.

The balanced chemical equation is

$$MgCO_3 \xrightarrow{\Delta} MgO + CO_2$$

Molar mass of $MgCO_3$ is 84 g mol^{-1}.

84 g $MgCO_3$ contain 24 g of Magnesium.

∴ 100 g of $MgCO_3$ contain

$$\frac{24 \text{ g Mg}}{84 \text{ g } MgCO_3} \times 100 \text{ g } MgCO_3$$

= 28.57 g Mg.

i.e. percentage of magnesium

= 28.57 %.

84 g $MgCO_3$ contain 12 g of carbon

$$MgCO_3$$

$$(24) + (12) + (3 \times 16) = 84 \text{ g mol}^{-1}$$

∴ 100 g $MgCO_3$ contain

$$\frac{12 \text{ g C}}{84 \text{ g } MgCO_3} \times 100 \text{ g } MgCO_3$$

= 14.29 g of carbon.

∴ Percentage of carbon

= 14.29 %.

84 g $MgCO_3$ contain 48 g of oxygen

∴ 100 g $MgCO_3$ contains

$$\frac{48 \text{ g O}}{84 \text{ g } MgCO_3} \times 100 \text{ g } MgCO_3$$

= 57.14 g of oxygen.

∴ Percentage of oxygen

= 57.14 %.

As per the stoichiometric equation,

84 g of 100 % pure $MgCO_3$ on heating gives 44 g of CO_2.

∴ 1000 g of 90 % pure $MgCO_3$ gives

$$= \frac{44 \text{ g}}{84 \text{ g}} \times 90\% \times 1000 \text{ g}$$

= 471.43 g of CO_2

= 0.471 kg of CO_2